

Summary Statistics of Numeric Columns				
Features	Min	Max	Avg	Range
Age	16	59	37	43
Listening Time (minutes)	10	299	154	289
Songs Played per Day	1	99	50	98
Skip Rate (%)	0	0.6	0.3	0.6
Ads Listened per Week	0	49	6.9	49

Features	Insights
Age	narrow (16–59) with no invalid values, indicating consistent demographic coverage across users
Listening Time (minutes)	Listening time spans a wide range (10–299), indicating high variability in user engagement levels and suitability for segmentation.
Songs played per day	Songs played per day is evenly distributed around the average (50), with a minimum of 1 indicating no inactive users in the dataset
Skip rate (%)	Skip rate ranges from 0 to 0.6 with an average of 0.3, indicating moderate variation in user satisfaction and consistent behavior across users
Ads listened per week	Ads listened per week ranges from 0 to 49 with an average of 6.9, indicating variability in ad exposure across users

Key Behavioral Features & Definitions	
Listening Time (minutes)	Represents the depth of user engagement based on total time spent listening.
Songs Played per Day	Represents the frequency of user activity and consistency of platform usage.
Skip Rate (%)	Represents listener satisfaction and listening intent, indicating how often content is skipped.
ads_listened_per_week & is_churned will not be considered a key behavioral feature because it provides context about a listener but doesn't define engagement levels of a listener.	

Listener Segmentation Behavioral Definitions		
Causal	Recurring	Retained
listeners with low engagement and high skipping behavior, indicating low retention potential	Listeners with consistent engagement and moderate activity, representing stable but not maximized usage.	Listeners with consistent engagement and moderate activity, representing stable but not maximized usage.

Listener Segmentation Thresholds			
Features	Below AVG	Around AVG	Above AVG
Listening Time (minutes)	<120	120-180	180>
Songs Played per Day	<40	40-65	65>
Skip Rate (%)	<0.2	0.2-.04	0.4>
listeners were segmented into low, medium, and high engagement groups using average-based thresholds across key behavioral metrics.			

During EDA, I evaluated all available features, including ads_listened_per_week. However, when defining user segments, I prioritized core behavioral metrics that directly reflect engagement—such as listening time, songs played per day, and skip rate.